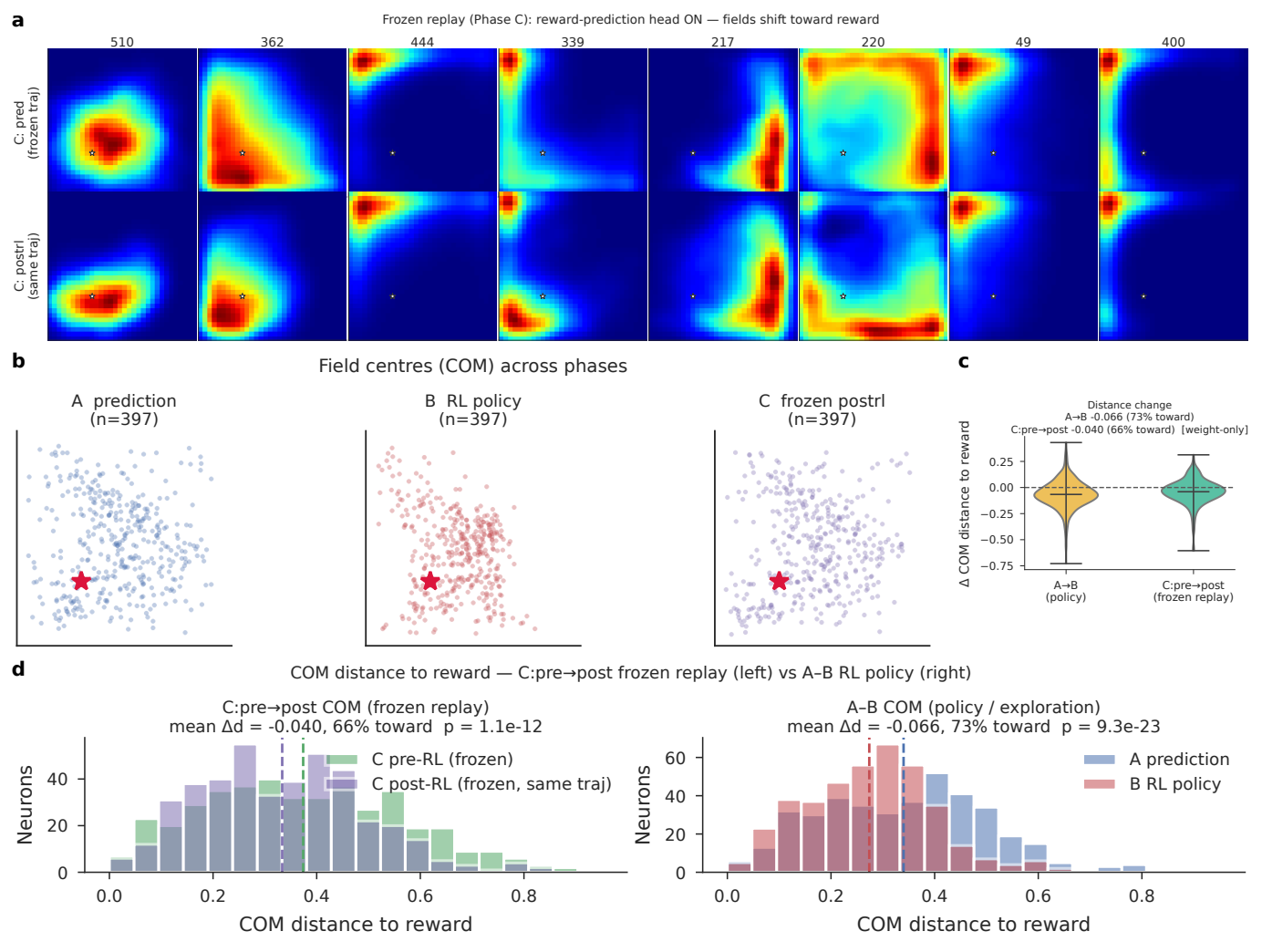

a
Frozen replay (Phase C): reward-prediction head ON — fields shift toward reward
510
362
444
339
217
220
49
400
C: pred (frozen traj)
C: postrl (same traj)
b
Field centres (COM) across phases
A prediction (n=397)
B RL policy (n=397)
C frozen postrl (n=397)
c
Distance change
A→B -0.066 (73% toward)
C:pre→post -0.040 (66% toward) [weight-only]
Δ COM distance to reward
0.25
0.00
−0.25
−0.50
−0.75
A→B (policy)
C:pre→post (frozen replay)
d
COM distance to reward — C:pre→post frozen replay (left) vs A–B RL policy (right)
C:pre→post COM (frozen replay)
mean Δd = -0.040, 66% toward p = 1.1e-12
C pre-RL (frozen)
C post-RL (frozen, same traj)
Neurons
COM distance to reward
A–B COM (policy / exploration)
mean Δd = -0.066, 73% toward p = 9.3e-23
A prediction
B RL policy
Neurons
COM distance to reward